GINS ESKF: IMU Propagation and GNSS Measurement Update

1 Purpose

This note connects two ways of deriving the ESKF state-transition matrix F:

1. derive the continuous-time error-state kinematics and then discretize them;
2. start directly from the discrete nominal-state propagation and linearize it with respect to local state perturbations.

They describe the same propagation of small errors. When the same approximation order is retained, the results agree. Direct differentiation of a discrete propagation may retain higher-order terms in Δt that are absent from a first-order discretization $F_d \approx I + F_c \Delta t$.

2 True, nominal, and error states

The true state is represented by a nominal state plus a small error. For Euclidean variables,

$$\mathbf{p}^t = \mathbf{p} + \delta\mathbf{p}, \tag{1}$$
$$\mathbf{v}^t = \mathbf{v} + \delta\mathbf{v}, \tag{2}$$
$$\mathbf{b}_\mathrm{g}{}^t = \mathbf{b}_\mathrm{g} + \delta\mathbf{b}_\mathrm{g}, \tag{3}$$
$$\mathbf{b}_\mathrm{a}{}^t = \mathbf{b}_\mathrm{a} + \delta\mathbf{b}_\mathrm{a}, \tag{4}$$
$$\mathbf{g}^t = \mathbf{g} + \delta\mathbf{g}. \tag{5}$$

For rotation, using a right perturbation,

$$\mathbf{R}^t = \mathbf{R}\,\mathrm{Exp}(\delta\boldsymbol{\theta}). \tag{6}$$

The 18-dimensional error state is

$$\delta\boldsymbol{x} = \begin{bmatrix}\delta\mathbf{p} & \delta\mathbf{v} & \delta\boldsymbol{\theta} & \delta\mathbf{b}_\mathrm{g} & \delta\mathbf{b}_\mathrm{a} & \delta\mathbf{g}\end{bmatrix}^T. \tag{7}$$

3 Two routes to the discrete transition matrix

The textbook route starts from continuous-time error dynamics,

$$\delta\dot{\boldsymbol{x}} = F_c\,\delta\boldsymbol{x} + G_c\boldsymbol{w}, \tag{8}$$

and then discretizes them. A first-order discretization gives

$$F_d \approx I + F_c\Delta t. \tag{9}$$

The direct discrete route first writes the nominal propagation

$$\boldsymbol{x}_{k+1} = f_d(\boldsymbol{x}_k, \boldsymbol{u}_k), \tag{10}$$

perturbs the input state, propagates it through the same discrete function, and measures the resulting output perturbation:

$$\boldsymbol{x}_k^t = \boldsymbol{x}_k \oplus \delta\boldsymbol{x}_k, \tag{11}$$
$$\boldsymbol{x}_{k+1}^t = f_d(\boldsymbol{x}_k \oplus \delta\boldsymbol{x}_k, \boldsymbol{u}_k), \tag{12}$$
$$\boldsymbol{x}_{k+1}^t = \boldsymbol{x}_{k+1} \oplus \delta\boldsymbol{x}_{k+1}. \tag{13}$$

Linearization gives

$$\delta\boldsymbol{x}_{k+1} \approx F_d\, \delta\boldsymbol{x}_k. \tag{14}$$

Thus F_d is the local Jacobian describing how the current error propagates into the next error.

4 Discrete nominal-state propagation

Using the Chapter 3 notation, the IMU measurements are denoted by $\tilde{\mathbf{a}}$ and $\tilde{\boldsymbol{\omega}}$. The increments used in propagation are

$$\boldsymbol{\tau} \triangleq (\tilde{\boldsymbol{\omega}} - \mathbf{b}_\mathrm{g})\Delta t. \tag{15}$$

The nominal propagation considered here is

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k\Delta t + \frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})\,\Delta t^2 + \frac{1}{2}\mathbf{g}_k\Delta t^2, \tag{16}$$
$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})\,\Delta t + \mathbf{g}_k\Delta t, \tag{17}$$
$$\mathbf{R}_{k+1} = \mathbf{R}_k\,\mathrm{Exp}(\boldsymbol{\tau}), \tag{18}$$
$$\mathbf{b}_{\mathrm{g}k+1} = \mathbf{b}_{\mathrm{g}k}, \tag{19}$$
$$\mathbf{b}_{\mathrm{a}k+1} = \mathbf{b}_{\mathrm{a}k}, \tag{20}$$
$$\mathbf{g}_{k+1} = \mathbf{g}_k. \tag{21}$$

5 Position row of F

Starting from

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k\Delta t + \frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})\Delta t^2 + \frac{1}{2}\mathbf{g}_k\Delta t^2, \tag{22}$$

we obtain immediately

$$F_{pp} = I, \tag{23}$$
$$F_{pv} = I\Delta t. \tag{24}$$

For the rotation term, apply a right perturbation:

$$\mathbf{R}_k^t = \mathbf{R}_k\,\mathrm{Exp}(\delta\boldsymbol{\theta}_k) \approx \mathbf{R}_k(I + \delta\boldsymbol{\theta}_k^\wedge). \tag{25}$$

Then

$$\mathbf{R}_k^t(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a}) \approx \mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a}) + \mathbf{R}_k\delta\boldsymbol{\theta}_k^\wedge(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a}) \tag{26}$$
$$= \mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a}) - \mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})^\wedge\,\delta\boldsymbol{\theta}_k, \tag{27}$$

where $\delta\boldsymbol{\theta}^\wedge(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a}) = -(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})^\wedge\delta\boldsymbol{\theta}$. Therefore

$$\boxed{F_{p\theta} = -\frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})^\wedge\Delta t^2.} \tag{28}$$

Because the accelerometer term is $\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a}$,

$$\boxed{F_{pb_a} = -\frac{1}{2}\mathbf{R}_k\Delta t^2.} \tag{29}$$

The gravity contribution is

$$\boxed{F_{pg} = \frac{1}{2}I\Delta t^2.} \tag{30}$$

There is no direct b_g dependence in this discrete position update, hence

$$F_{pb_g} = 0. \tag{31}$$

The complete position error propagation is therefore

$$\delta\mathbf{p}_{k+1} = \delta\mathbf{p}_k + \Delta t\,\delta\mathbf{v}_k - \frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})^\wedge\Delta t^2\delta\boldsymbol{\theta}_k \tag{32}$$

$$- \frac{1}{2}\mathbf{R}_k\Delta t^2\delta\mathbf{b}_{\mathrm{a}k} + \frac{1}{2}I\Delta t^2\delta\mathbf{g}_k. \tag{33}$$

6 Velocity row of F

From

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})\Delta t + \mathbf{g}_k\Delta t, \tag{34}$$

we obtain

$$F_{vp} = 0, \tag{35}$$

$$F_{vv} = I, \tag{36}$$

$$\boxed{F_{v\theta}} = \boxed{-\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})^\wedge\Delta t}, \tag{37}$$

$$F_{vb_g} = 0, \tag{38}$$

$$\boxed{F_{vb_a}} = \boxed{-\mathbf{R}_k\Delta t}, \tag{39}$$

$$\boxed{F_{vg}} = \boxed{I\Delta t}. \tag{40}$$

Thus

$$\delta\mathbf{v}_{k+1} = \delta\mathbf{v}_k - \mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_\mathrm{a})^\wedge\Delta t\,\delta\boldsymbol{\theta}_k - \mathbf{R}_k\Delta t\,\delta\mathbf{b}_{\mathrm{a}k} + I\Delta t\,\delta\mathbf{g}_k. \tag{41}$$

7 Rotation row of F and the manif Jacobians

The rotation propagation is

$$\mathbf{R}_{k+1} = \mathbf{R}_k \oplus \boldsymbol{\tau} = \mathbf{R}_k\,\mathrm{Exp}(\boldsymbol{\tau}), \qquad \boldsymbol{\tau} = (\tilde{\boldsymbol{\omega}} - \mathbf{b}_\mathrm{g})\Delta t. \tag{42}$$

A `manif` call of the form

```
new_R = R_.rplus(tau, J_R, J_tau);
```

returns the propagated rotation together with the local Jacobians

$$J_R = \frac{\partial(\mathbf{R} \oplus \boldsymbol{\tau})}{\partial \mathbf{R}}, \tag{43}$$

$$J_\tau = \frac{\partial(\mathbf{R} \oplus \boldsymbol{\tau})}{\partial \boldsymbol{\tau}}. \tag{44}$$

For the right-plus convention on $SO(3)$,

$$\boxed{J_R = \mathrm{Exp}(-\boldsymbol{\tau})}, \tag{45}$$

$$\boxed{J_\tau = J_r(\boldsymbol{\tau})}. \tag{46}$$

Therefore

$$\boxed{F_{\theta\theta} = J_R = \mathrm{Exp}(-\boldsymbol{\tau}).} \tag{47}$$

Since

$$\frac{\partial \boldsymbol{\tau}}{\partial \mathbf{b}_\mathrm{g}} = -I\Delta t, \tag{48}$$

the chain rule gives

$$\boxed{F_{\theta b_g} = -J_\tau \Delta t = -J_r(\boldsymbol{\tau})\Delta t.} \tag{49}$$

Hence

$$\delta\boldsymbol{\theta}_{k+1} = J_R \delta\boldsymbol{\theta}_k - J_\tau \Delta t \, \delta\mathbf{b}_{\mathrm{g}k}. \tag{50}$$

For a small IMU interval,

$$J_R = \mathrm{Exp}(-\boldsymbol{\tau}) \approx I - (\tilde{\boldsymbol{\omega}} - \mathbf{b}_\mathrm{g})^\wedge \Delta t + O(\Delta t^2), \tag{51}$$

$$J_r(\boldsymbol{\tau}) \approx I - \frac{1}{2}\boldsymbol{\tau}^\wedge + O(\|\boldsymbol{\tau}\|^2). \tag{52}$$

Consequently

$$-J_r(\boldsymbol{\tau})\Delta t = -I\Delta t + O(\Delta t^2), \tag{53}$$

which explains the textbook first-order approximation $F_{\theta b_g} \approx -I\Delta t$.

8 Bias and gravity rows

With random-walk driving noise omitted from the deterministic state-transition Jacobian,

$$\delta\mathbf{b}_{\mathrm{g}k+1} = \delta\mathbf{b}_{\mathrm{g}k}, \tag{54}$$

$$\delta\mathbf{b}_{\mathrm{a}k+1} = \delta\mathbf{b}_{\mathrm{a}k}, \tag{55}$$

$$\delta\mathbf{g}_{k+1} = \delta\mathbf{g}_k. \tag{56}$$

Thus

$$F_{b_g b_g} = I, \qquad F_{b_a b_a} = I, \qquad F_{gg} = I, \tag{57}$$

and their other deterministic blocks are zero.

9 Complete direct-discrete state-transition matrix

Using

$$\boldsymbol{\tau} = (\tilde{\boldsymbol{\omega}} - \mathbf{b}_{\mathrm{g}})\Delta t, \tag{58}$$

the direct Jacobian of the stated discrete nominal propagation is

$$\boxed{F_d = \begin{bmatrix} I & I\Delta t & -\frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_{\mathrm{a}})^{\wedge}\Delta t^2 & 0 & -\frac{1}{2}\mathbf{R}_k\Delta t^2 & \frac{1}{2}I\Delta t^2 \\ 0 & I & -\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_{\mathrm{a}})^{\wedge}\Delta t & 0 & -\mathbf{R}_k\Delta t & I\Delta t \\ 0 & 0 & J_R & -J_\tau\Delta t & 0 & 0 \\ 0 & 0 & 0 & I & 0 & 0 \\ 0 & 0 & 0 & 0 & I & 0 \\ 0 & 0 & 0 & 0 & 0 & I \end{bmatrix}}, \tag{59}$$

where

$$J_R = \mathrm{Exp}(-\boldsymbol{\tau}), \qquad J_\tau = J_r(\boldsymbol{\tau}). \tag{60}$$

10 Why the textbook position row is simpler

For the continuous error-state system, the position equation is

$$\delta\dot{\mathbf{p}} = \delta\mathbf{v}. \tag{61}$$

Therefore the continuous-time block from rotation error to position-error rate is zero:

$$(F_c)_{p\theta} = 0. \tag{62}$$

If the discrete transition is approximated only to first order,

$$F_d \approx I + F_c\Delta t, \tag{63}$$

then

$$(F_d)_{p\theta} = 0. \tag{64}$$

The direct discrete position equation, however, contains the term

$$\frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_{\mathrm{a}})\Delta t^2, \tag{65}$$

so its Jacobian contains

$$F_{p\theta} = -\frac{1}{2}\mathbf{R}_k(\tilde{\mathbf{a}} - \mathbf{b}_{\mathrm{a}})^{\wedge}\Delta t^2. \tag{66}$$

This is a second-order term. It is not a contradiction: it is simply discarded by the first-order approximation $I + F_c\Delta t$. A higher-order expansion

$$e^{F_c\Delta t} = I + F_c\Delta t + \frac{1}{2}F_c^2\Delta t^2 + \cdots \tag{67}$$

recovers such indirect couplings, e.g. rotation error $\rightarrow$ velocity error $\rightarrow$ position error.

Similarly,

$$J_R = \mathrm{Exp}(-(\tilde{\boldsymbol{\omega}} - \mathbf{b}_{\mathrm{g}})\Delta t) = I - (\tilde{\boldsymbol{\omega}} - \mathbf{b}_{\mathrm{g}})^{\wedge}\Delta t + O(\Delta t^2), \tag{68}$$

so the exact/local discrete Lie-group Jacobian and the textbook first-order rotation block agree to first order.

11 Main interpretation

The essential quantity is not "the nominal-state Jacobian" versus "the error-state Jacobian" as two unrelated objects. The transition matrix F_d describes how a small state error before propagation becomes a small state error after propagation:

$$\boxed{\delta \boldsymbol{x}_{k+1} \approx F_d \, \delta \boldsymbol{x}_k.} \tag{69}$$

There are two useful derivation routes:

$$\text{continuous route: true kinematics} \to \text{continuous error dynamics} \to F_c \to F_d,$$
$$\text{discrete route: discrete nominal propagation} \to \text{local perturbation} \to F_d.$$

The `manif` Jacobians implement the second viewpoint naturally for Lie-group variables such as $R \in SO(3)$. For Euclidean variables such as p and v, the same idea reduces to ordinary differentiation.

12 Implementation consistency note

If the discrete equations for p_{k+1} and v_{k+1} use the pre-propagation rotation R_k, their direct Jacobians should also be evaluated at R_k. Updating the stored rotation to R_{k+1} before constructing those Jacobian blocks changes the linearization point and is not exactly the Jacobian of the discrete equations written above.

13 From IMU Propagation to GNSS Correction

The IMU propagation developed above produces a predicted nominal state and a predicted covariance. In a GINS ESKF, GNSS measurements provide the complementary correction step. The overall cycle is

$$\boxed{\text{IMU propagation} \longrightarrow (\boldsymbol{x}^-, P^-) \longrightarrow \text{GNSS measurement update} \longrightarrow (\boldsymbol{x}^+, P^+).} \tag{70}$$

The superscript "$-$" denotes the state before a measurement correction, while "$+$" denotes the corrected state.

14 GNSS Initialization

The implementation treats the first GNSS observation differently from later observations. Instead of forming a Kalman innovation, the first valid GNSS pose establishes the initial nominal pose:

$$\mathbf{R} \leftarrow \mathbf{R}_{\text{GNSS}}, \tag{71}$$
$$\mathbf{p} \leftarrow \mathbf{p}_{\text{GNSS}}. \tag{72}$$

This provides a reasonable initial reference for the ESKF before small-error linearization is used. The first GNSS observation therefore acts as an initialization step rather than an ordinary measurement correction.

In the implementation, this step initializes only position and orientation. Velocity, IMU biases, gravity, and covariance are not overwritten by this GNSS initialization. The filter time is also moved to the GNSS timestamp.

This distinction is important because an ESKF assumes that the error state is small. If the nominal position were still near an arbitrary origin while the GNSS position were already expressed in a large global coordinate system, the initial innovation could be far outside the small-error regime.

15 GNSS Pose Measurement Model

The GNSS update can be derived with exactly the same linearization idea used for IMU propagation. The difference is the function being linearized.

For prediction, the nonlinear function is the motion model,

$$\boldsymbol{x}_{k+1} = f_d(\boldsymbol{x}_k, \boldsymbol{u}_k), \tag{73}$$

and linearizing the propagated output error gives the state-transition Jacobian F.

For measurement update, the nonlinear function is the measurement model,

$$\boldsymbol{z}_k = h(\boldsymbol{x}_k) + \boldsymbol{n}_k, \tag{74}$$

and linearizing the measurement residual with respect to the error state gives the measurement Jacobian H.

Thus, H is *not* obtained by differentiating the Kalman correction equation. It is obtained by perturbing the state inside the measurement model and observing how the predicted measurement changes.

For the implementation considered here, GNSS provides a six-dimensional pose observation: position and orientation. The nominal measurement prediction is therefore

$$h(\boldsymbol{x}) = \begin{bmatrix} \mathbf{p} \\ \mathbf{R} \end{bmatrix}. \tag{75}$$

Because the second component lies on $SO(3)$, the measurement residual cannot be formed by ordinary subtraction for all six components. Position uses Euclidean subtraction, while orientation uses the $SO(3)$ group difference.

The error-state ordering remains

$$\delta\boldsymbol{x} = \begin{bmatrix} \delta\mathbf{p} & \delta\mathbf{v} & \delta\boldsymbol{\theta} & \delta\mathbf{b}_{\mathrm{g}} & \delta\mathbf{b}_{\mathrm{a}} & \delta\mathbf{g} \end{bmatrix}^T. \tag{76}$$

16 Deriving the GNSS Measurement Jacobian H

We now derive H from the measurement equations, in the same spirit as the direct discrete derivation of F.

16.1 Position measurement

Let the true position be

$$\mathbf{p}^t = \mathbf{p} + \delta\mathbf{p}. \tag{77}$$

The GNSS position observation is modeled as

$$\mathbf{p}_{\mathrm{obs}} = \mathbf{p}^t + \boldsymbol{n}_p. \tag{78}$$

Substituting the error-state definition gives

$$\mathbf{p}_{\mathrm{obs}} = \mathbf{p} + \delta\mathbf{p} + \boldsymbol{n}_p. \tag{79}$$

The nominal predicted measurement is

$$\hat{\mathbf{p}}_{\mathrm{obs}} = h_p(\boldsymbol{x}) = \mathbf{p}. \tag{80}$$

Therefore the position residual is

$$\boldsymbol{r}_p = \mathbf{p}_{\text{obs}} - \mathbf{p} \tag{81}$$
$$= \delta\mathbf{p} + \boldsymbol{n}_p. \tag{82}$$

Hence, with respect to the complete 18-dimensional error state,

$$\boxed{H_p = \begin{bmatrix} I & 0 & 0 & 0 & 0 & 0 \end{bmatrix}.} \tag{83}$$

Equivalently,

$$\frac{\partial \boldsymbol{r}_p}{\partial \delta\mathbf{p}} = I, \tag{84}$$

while its derivatives with respect to $\delta\mathbf{v}$, $\delta\boldsymbol{\theta}$, $\delta\mathbf{b}_{\text{g}}$, $\delta\mathbf{b}_{\text{a}}$, and $\delta\mathbf{g}$ are zero.

16.2 Orientation measurement

For rotation, the true attitude is represented by the right perturbation

$$\mathbf{R}^t = \mathbf{R} \oplus \delta\boldsymbol{\theta} = \mathbf{R}\,\text{Exp}(\delta\boldsymbol{\theta}). \tag{85}$$

The nominal predicted orientation is $\mathbf{R}$, and the local rotational residual is defined by the right-minus operation

$$\boldsymbol{r}_R = \mathbf{R}_{\text{obs}} \ominus \mathbf{R} = \text{Log}\big(\mathbf{R}^{-1}\mathbf{R}_{\text{obs}}\big). \tag{86}$$

To derive the measurement Jacobian, we linearize the residual with respect to the *state error*. As in the standard EKF derivation, the measurement Jacobian is evaluated at zero measurement noise; the measurement noise is handled separately through the measurement covariance. Hence, for the purpose of computing H_R, set $\boldsymbol{n}_R = 0$, so that the ideal observation is the true attitude,

$$\mathbf{R}_{\text{obs}} = \mathbf{R}^t. \tag{87}$$

Using the right-plus/right-minus definitions from Micro Lie Theory,

$$\boldsymbol{r}_R(\delta\boldsymbol{\theta}, 0) = \mathbf{R}^t \ominus \mathbf{R} \tag{88}$$
$$= (\mathbf{R} \oplus \delta\boldsymbol{\theta}) \ominus \mathbf{R} \tag{89}$$
$$= \delta\boldsymbol{\theta}. \tag{90}$$

The last equality is the exact local inverse identity

$$(X \oplus \boldsymbol{\tau}) \ominus X = \boldsymbol{\tau}, \tag{91}$$

so no BCH approximation is required. Therefore,

$$\frac{\partial \boldsymbol{r}_R}{\partial \delta\boldsymbol{\theta}} = I, \tag{92}$$

and with respect to the complete 18-dimensional error state,

$$\boxed{H_R = \begin{bmatrix} 0 & 0 & I & 0 & 0 & 0 \end{bmatrix}.} \tag{93}$$

The rotational measurement noise is still present in the stochastic measurement model and contributes to the measurement covariance V. If one keeps a finite nonzero $\boldsymbol{n}_R$ inside the group composition, the residual generally does not reduce exactly to $\delta\boldsymbol{\theta} + \boldsymbol{n}_R$; that simplification would require a first-order approximation. Such an approximation is unnecessary for deriving H_R because H_R is the derivative with respect to the state error evaluated at zero measurement noise.

16.3 Complete pose measurement Jacobian

Stacking the position and orientation residuals,

$$\boldsymbol{r} = \begin{bmatrix} \boldsymbol{r}_p \\ \boldsymbol{r}_R \end{bmatrix}, \tag{94}$$

we obtain, to first order,

$$\boldsymbol{r} \approx H\,\delta\boldsymbol{x} + \boldsymbol{n}, \tag{95}$$

with

$$\boxed{H = \begin{bmatrix} I & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & I & 0 & 0 & 0 \end{bmatrix}.} \tag{96}$$

This is exactly the matrix assembled in the implementation:

```
H.block<3, 3>(0, 0) = Mat3d::Identity();  // position residual wrt dp
H.block<3, 3>(3, 6) = Mat3d::Identity();  // rotation residual wrt dtheta
```

17 GNSS Innovation

The derivation above also explains the innovation used by the code. The position innovation is the ordinary Euclidean difference

$$\boxed{\boldsymbol{r}_p = \mathbf{p}_{\text{obs}} - \mathbf{p}.} \tag{97}$$

For orientation, the innovation is the local group difference

$$\boxed{\boldsymbol{r}_R = \operatorname{Log}\!\left(\mathbf{R}^{-1}\mathbf{R}_{\text{obs}}\right).} \tag{98}$$

The complete six-dimensional innovation is therefore

$$\boxed{\boldsymbol{r} = \begin{bmatrix} \mathbf{p}_{\text{obs}} - \mathbf{p} \\ \operatorname{Log}\!\left(\mathbf{R}^{-1}\mathbf{R}_{\text{obs}}\right) \end{bmatrix}.} \tag{99}$$

The `manif` implementation is

```
innov.head<3>() = pose.translation() - p_;
innov.tail<3>() = R_obs.rminus(R_).coeffs();
```

where

$$\mathbf{R}_{\text{obs}} \ominus \mathbf{R} = \operatorname{Log}(\mathbf{R}^{-1}\mathbf{R}_{\text{obs}}). \tag{100}$$

Thus the prediction and measurement linearizations have the same conceptual structure:

$$\text{Predict:} \quad \delta\boldsymbol{x}_{k+1} \approx F\,\delta\boldsymbol{x}_k, \tag{101}$$

$$\text{Update:} \quad \boldsymbol{r}_k \approx H\,\delta\boldsymbol{x}_k + \boldsymbol{n}_k. \tag{102}$$

The difference is simply the output being linearized: F maps state error to the next state error, whereas H maps state error to measurement residual.

18 Innovation Covariance and Kalman Gain

Let the GNSS measurement-noise covariance be V. In the implementation it is a 6×6 diagonal matrix containing horizontal position variance, height variance, and three orientation variances. The innovation covariance is

$$\boxed{S = HP^{-}H^{T} + V.} \tag{103}$$

The Kalman gain is

$$\boxed{K = P^{-}H^{T}S^{-1}.} \tag{104}$$

The estimated error state is then

$$\boxed{\widehat{\delta \boldsymbol{x}} = K\boldsymbol{r}.} \tag{105}$$

This is a key ESKF distinction: the Kalman update estimates the *error state*; it does not replace the nominal state directly.

19 Covariance Update Before Error Injection

The implementation uses the standard simplified covariance update

$$\boxed{P \leftarrow (I - KH)P^{-}.} \tag{106}$$

A numerically more robust alternative is the Joseph form

$$P \leftarrow (I - KH)P^{-}(I - KH)^{T} + KVK^{T}, \tag{107}$$

but the equations in this section follow the implementation directly.

20 Wheel Velocity Measurement Update: Body and World Frames

We now assume a sensor provides the full three-dimensional velocity $\boldsymbol{z}_v^B \in \mathbb{R}^3$ expressed in the body frame. This is a modeling assumption for the derivation, not a claim that an ordinary wheel encoder measures all three components. We neglect lever arms and additional motion constraints. The nominal velocity $\mathbf{v}$ is expressed in the world frame, and $\mathbf{R}$ maps body-frame vectors to the world frame.

20.1 Body-frame observation model

The physical measurement equation is

$$\boxed{\boldsymbol{z}_v^B = \mathbf{R}^{T}\mathbf{v} + \boldsymbol{n}_v^B, \qquad h_B(\boldsymbol{x}) = \mathbf{R}^{T}\mathbf{v}.} \tag{108}$$

Its innovation, evaluated at the predicted nominal state, is

$$\boldsymbol{r}_B = \boldsymbol{z}_v^B - \mathbf{R}^{T}\mathbf{v}. \tag{109}$$

We use the same right-local attitude error as in the preceding sections:

$$\mathbf{R}^{t} = \mathbf{R}\,\mathrm{Exp}(\delta\boldsymbol{\theta}), \qquad \mathbf{v}^{t} = \mathbf{v} + \delta\boldsymbol{v}. \tag{110}$$

Consequently, the noise-free observation at the perturbed state is

$$h_B(\boldsymbol{x} \oplus \delta\boldsymbol{x}) = (\mathbf{R}\,\mathrm{Exp}(\delta\boldsymbol{\theta}))^T(\mathbf{v} + \delta\boldsymbol{v}) \tag{111}$$

$$= \mathrm{Exp}(-\delta\boldsymbol{\theta})\mathbf{R}^T(\mathbf{v} + \delta\boldsymbol{v}). \tag{112}$$

Write $\boldsymbol{q} = \mathbf{R}^T\mathbf{v}$. The derivative with respect to the velocity perturbation is exact:

$$\left.\frac{\partial h_B}{\partial \delta\boldsymbol{v}}\right|_0 = \mathbf{R}^T. \tag{113}$$

For attitude, use the defining derivative of the exponential at the origin, $\frac{d}{d\epsilon}\,\mathrm{Exp}(-\epsilon\boldsymbol{u})\big|_{\epsilon=0} = -\boldsymbol{u}^\wedge$, rather than a BCH expansion:

$$\left.\frac{d}{d\epsilon}\,\mathrm{Exp}(-\epsilon\boldsymbol{u})\boldsymbol{q}\right|_{\epsilon=0} = -\boldsymbol{u}^\wedge\boldsymbol{q} = \boldsymbol{q}^\wedge\boldsymbol{u}. \tag{114}$$

Therefore the local measurement Jacobian is

$$\boxed{H_B = \begin{bmatrix} 0_{3\times3} & \mathbf{R}^T & \mathbf{R}^T\mathbf{v}^\wedge & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \end{bmatrix}.} \tag{115}$$

This derivative is exact at the nominal state; the linearized measurement relation $\boldsymbol{r}_B \simeq H_B\delta\boldsymbol{x} + \boldsymbol{n}_v^B$ is a first-order approximation for finite perturbations. The update uses the covariance $V_B = \mathrm{Cov}(\boldsymbol{n}_v^B)$ in the body frame.

20.2 World-frame observation used in Chapter 3

Chapter 3, Eqs. (3.74)–(3.75), instead first rotates the measured velocity with the *estimated* attitude $\widehat{\mathbf{R}}$:

$$\boldsymbol{z}_v^W = \widehat{\mathbf{R}}\boldsymbol{z}_v^B, \qquad \boldsymbol{r}_W = \boldsymbol{z}_v^W - \mathbf{v}. \tag{116}$$

Treating $\boldsymbol{z}_v^W$ as fixed when differentiating, the abstract observation model is $h_W(\boldsymbol{x}) = \mathbf{v}$. Perturbing the state yields

$$h_W(\boldsymbol{x} \oplus \delta\boldsymbol{x}) = \mathbf{v} + \delta\boldsymbol{v}, \tag{117}$$

so that

$$\boxed{H_W = \begin{bmatrix} 0_{3\times3} & I_3 & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} & 0_{3\times3} \end{bmatrix}.} \tag{118}$$

This reproduces Eq. (3.75) and the simplified Jacobian used by `ObserveWheelSpeed()`. If V_B is the body-frame sensor covariance, rotating only the sensor noise gives

$$V_W = \widehat{\mathbf{R}}V_B\widehat{\mathbf{R}}^T. \tag{119}$$

However, $\widehat{\mathbf{R}}$ is itself estimated from the filter state. The fixed-transformed-measurement formulation omits its state dependence and associated correlations; the two Jacobians should therefore not be interpreted as statistically identical observation models.

20.3 Comparison and Kalman update

	Body frame	World frame (Chapter 3)
Observation h	$\mathbf{R}^T\mathbf{v}$	$\mathbf{v}$
Innovation	$\boldsymbol{z}_v^B - \mathbf{R}^T\mathbf{v}$	$\widehat{\mathbf{R}}\boldsymbol{z}_v^B - \mathbf{v}$
Velocity block	$\mathbf{R}^T$	I_3
Attitude block	$\mathbf{R}^T\mathbf{v}^\wedge$	$0_{3\times3}$

For either chosen model, with the matching residual $\boldsymbol{r}$, Jacobian H, and measurement covariance V, the ESKF update is

$$S = HP^{-}H^{T} + V, \tag{120}$$

$$K = P^{-}H^{T}S^{-1}, \tag{121}$$

$$\widehat{\delta\boldsymbol{x}} = K\boldsymbol{r}. \tag{122}$$

The subsequent covariance update, error injection, and reset are the same operations as in the GNSS update. The difference lies in how the velocity measurement and its Jacobian are modeled.

21 Injecting the Estimated Error into the Nominal State

Write the estimated correction as

$$\widehat{\delta\boldsymbol{x}} = \begin{bmatrix} \widehat{\delta\mathbf{p}} & \widehat{\delta\mathbf{v}} & \widehat{\delta\boldsymbol{\theta}} & \widehat{\delta\mathbf{b}_{\mathrm{g}}} & \widehat{\delta\mathbf{b}_{\mathrm{a}}} & \widehat{\delta\mathbf{g}} \end{bmatrix}^{T}. \tag{123}$$

It is convenient to regard the complete nominal state as the product manifold

$$\mathcal{X} = \mathbb{R}^3 \times \mathbb{R}^3 \times SO(3) \times \mathbb{R}^3 \times \mathbb{R}^3 \times \mathbb{R}^3, \tag{124}$$

with

$$\boldsymbol{x} = (\mathbf{p}, \mathbf{v}, \mathbf{R}, \mathbf{b}_{\mathrm{g}}, \mathbf{b}_{\mathrm{a}}, \mathbf{g}). \tag{125}$$

Then the entire ESKF injection can be written compactly as

$$\boxed{\boldsymbol{x}^{+} = \boldsymbol{x}^{-} \oplus \widehat{\delta\boldsymbol{x}}.} \tag{126}$$

Here the composite $\oplus$ operation acts componentwise: it is ordinary vector addition on the Euclidean components and the right-plus operation on the $SO(3)$ component. Expanding the definition once gives

$$\mathbf{p}^{+} = \mathbf{p}^{-} + \widehat{\delta\mathbf{p}}, \tag{127}$$

$$\mathbf{v}^{+} = \mathbf{v}^{-} + \widehat{\delta\mathbf{v}}, \tag{128}$$

$$\mathbf{R}^{+} = \mathbf{R}^{-} \oplus \widehat{\delta\boldsymbol{\theta}}, \tag{129}$$

$$\mathbf{b}_{\mathrm{g}}^{+} = \mathbf{b}_{\mathrm{g}}^{-} + \widehat{\delta\mathbf{b}_{\mathrm{g}}}, \tag{130}$$

$$\mathbf{b}_{\mathrm{a}}^{+} = \mathbf{b}_{\mathrm{a}}^{-} + \widehat{\delta\mathbf{b}_{\mathrm{a}}}, \tag{131}$$

$$\mathbf{g}^{+} = \mathbf{g}^{-} + \widehat{\delta\mathbf{g}}. \tag{132}$$

For the rotational component,

$$\mathbf{R}^{-} \oplus \widehat{\delta\boldsymbol{\theta}} = \mathbf{R}^{-}\,\mathrm{Exp}(\widehat{\delta\boldsymbol{\theta}}), \tag{133}$$

so the implementation uses `manif::rplus()` directly:

```
R_ = R_.rplus(manif::SO3Tangentd(dx_.segment<3>(6)));
```

The current implementation stores the state components separately rather than as one composite `manif` object, so the Euclidean components are still updated explicitly in C++. Mathematically, however, all six updates are one operation: $\boldsymbol{x}^{+} = \boldsymbol{x}^{-} \oplus \widehat{\delta\boldsymbol{x}}$.

The gyro- and accelerometer-bias corrections are conditional on the corresponding configuration flags; gravity is updated unconditionally.

22 Error-State Reset and Covariance Projection

After the estimated error has been injected into the nominal state, the error-state mean must be reset to zero:

$$\boxed{\delta \boldsymbol{x} \leftarrow 0.} \tag{134}$$

For Euclidean components, this reset is only a change of reference. For rotation, however, the local tangent coordinates are attached to the newly corrected nominal orientation. Therefore the covariance must also be projected into the new error coordinates.

The implementation uses the reset Jacobian

$$J_{\text{reset}} = I_{18}, \tag{135}$$

except for the attitude block

$$\boxed{(J_{\text{reset}})_{\theta\theta} \approx I - \frac{1}{2}\widehat{\delta\boldsymbol{\theta}}^{\wedge}.} \tag{136}$$

The covariance is then transformed as

$$\boxed{P \leftarrow J_{\text{reset}} P J_{\text{reset}}^{T}.} \tag{137}$$

Only after this projection is the stored error-state vector set to zero.

23 Complete GINS ESKF Cycle

The complete IMU-prediction and GNSS-correction sequence can now be summarized as follows:

1. **IMU nominal propagation:** propagate $\mathbf{p}$, $\mathbf{v}$, and $\mathbf{R}$ using the IMU measurements; keep the nominal $\mathbf{b}_{\text{g}}$, $\mathbf{b}_{\text{a}}$, and $\mathbf{g}$ constant during propagation.
2. **Error covariance propagation:** construct the discrete transition matrix F and compute

$$P^{-} = F P^{+}_{\text{previous}} F^{T} + Q. \tag{138}$$

3. **GNSS innovation:** compare the observed GNSS pose with the predicted nominal pose using Euclidean subtraction for position and $SO(3)$ `rminus()` for orientation.
4. **Kalman correction:** compute S, K, and the estimated error state $\widehat{\delta\boldsymbol{x}}$.
5. **Covariance measurement update:** apply $(I - KH)P^{-}$ in the current implementation.
6. **Error injection:** inject $\widehat{\delta\boldsymbol{x}}$ into the nominal state, using `rplus()` for orientation.
7. **Reset:** project the covariance to the new local rotation coordinates and reset the error-state mean to zero.

Conceptually,

$$\boxed{\begin{array}{c} \text{IMU} \longrightarrow \text{Predict} \longrightarrow (\boldsymbol{x}^{-}, P^{-}) \\ \text{GNSS} \longrightarrow \text{innovation} \longrightarrow K\boldsymbol{r} \longrightarrow \text{inject} \longrightarrow \text{reset} \longrightarrow (\boldsymbol{x}^{+}, P^{+}). \end{array}} \tag{139}$$

24 Mapping to the ESKF Implementation

The main functions in the Chapter 3 implementation correspond to the mathematics above as follows:

Function	Role
`Predict()`	IMU nominal-state propagation, discrete error-state transition matrix F, and covariance propagation.
`ObserveGps()`	Uses the first GNSS pose for initialization; later measurements are forwarded to the pose update.
`ObserveSE3()`	Builds H, forms position and rotation innovations, computes K, estimates $\delta\boldsymbol{x}$, and performs the measurement covariance update.
`UpdateAndReset()`	Injects the estimated error into p, v, R, b_g, b_a, g, calls covariance projection, then resets the stored error state.
`ProjectCov()`	Applies the first-order attitude reset Jacobian to transform covariance into the new local error coordinates.

The two `manif` operations now have complementary roles:

$$\texttt{rplus}: \quad \mathbf{R} \oplus \delta\boldsymbol{\theta} = \mathbf{R}\,\mathrm{Exp}(\delta\boldsymbol{\theta}), \tag{140}$$

$$\texttt{rminus}: \quad \mathbf{R}_{\mathrm{obs}} \ominus \mathbf{R} = \mathrm{Log}(\mathbf{R}^{-1}\mathbf{R}_{\mathrm{obs}}). \tag{141}$$

Thus the same Lie-group perturbation convention is used consistently in both propagation and measurement correction.